GROUP ID: TY11-14

- Aditya malode -33339
- Ronak Shah -33368
- Aditya Yadav -33382
- Aditya Zalte -33383

Title of the project:ML Based Ship Route Prediction Using Ocean and Weather Data

- **Problem Definition**:

Ships currently use static sea charts, manual weather advice, and the master's/officer's experience to plan routes. This can lead to:

- Poor routes: Ignore changing sea conditions, causing longer travel times, more fuel use, and higher emissions.
- Wrong ETAs: Can cause port congestion, poor berth planning, and supply delays.
- Safety risks: Ships may enter rough-weather areas that could have been avoided with better route prediction.

- **Goal:**

- Predict the ship's future path (next latitude/longitude points).
- Predict the ETA at the next waypoint or port using current and forecast weather.
- Identify risky route areas where bad weather may force the ship to change its route.

Sponsoring Industry Details

- Name of Industry:ION DIGITAL
- Name of Industry Person:Maheshwar Bhosale
- Contact no. of industry person:90675 07393
- Email id of industry person:maheshwarcbhosale18@gmail.com

Project Flow

1 Data Collection



- Latitude & Longitude
- Timestamp
- Speed (SOG)
- Course (COG)
- Wind Speed & Direction
- Temperature
- Pressure
- Wave Height & Direction
- Ocean Current
- Sea Surface Temp

2 Data Preprocessing



- Remove duplicates
- Handle missing values
- Remove invalid locations
- Convert timestamp format
- Normalize numerical data

3 Data Integration



- Match data using time and location
- Merge AIS, weather and ocean data
- Create unified dataset

4 Feature Engineering



- Ship features (lat, lon, speed, course)
- Weather features (wind, temp, pressure)
- Ocean features (wave height, current speed, direction)
- Additional features (distance, change in speed/direction, time diff)

5 Sequence Creation



- Convert data into time-series
- Use past N time steps
- Each sequence contains ship, weather and ocean features

6 Dataset Splitting



- Training set – 70%
- Validation set – 15%
- Testing set – 15%

7 Model Training



LSTM Model

- Learns temporal patterns
- Uses sequential data
- Predicts future positions

8 Ship Route Prediction



- Predict next latitude
- Predict next longitude
- Generate future route

9 Performance Evaluation



- MAE (Mean Absolute Error)
- RMSE (Root Mean Square Error)
- Distance Error
- Route Prediction Accuracy

10 Final Output



- Predicted ship route
- Future ship locations
- Weather impact information
- Ocean condition impact information
- Potential navigation risks

Dataset Description

Name of Dataset	Source of Dataset	Characteristics of Dataset
AIS Data Complete + Weather	Kaggle — nadaemad2002	• AIS vessel positions with weather information • Includes speed, course and environmental parameters • Useful for trajectory prediction
2024 US Coast Guard + NOAA AIS Dataset	Kaggle (bwandowando) / originally NOAA Marine Cadastre	• Large-scale real-world AIS data • Contains vessel position, speed, course and vessel details • Useful for learning realistic vessel movement patterns
NOAA ICOADS	Kaggle mirror of NOAA International Comprehensive Ocean-Atmosphere Data Set	• Long-term ocean and weather observations • Includes wind, pressure, temperature and humidity • Useful for weather and ocean-condition analysis
AIS Ship Tracking – Vessel Dynamics & ETA Data	Kaggle — satyamrajput7913	• Vessel movement and dynamic information • Includes speed, course and ETA-related data • Useful for ETA prediction and benchmarking

Proposed AI/ML Model to solve the problem with Justification

Proposed approach — a hybrid multi-model pipeline, not a single model:

1. Trajectory prediction: Bi-directional LSTM / Seq2Seq with attention (or a lightweight Transformer encoder) over the AIS position sequence, with weather features concatenated at each timestep.
2. ETA prediction: Gradient-boosted trees (XGBoost/LightGBM) using engineered features (distance-to-go, current SOG, weather-severity index, historical route delay patterns).
3. Route pattern discovery / anomaly flagging: DBSCAN or HDBSCAN on historical trajectories to identify common shipping lanes and detect deviations caused by weather.

Justification

1. AIS trajectories are sequential and highly autocorrelated → recurrent/attention-based sequence models are the natural fit, unlike simple regression which ignores time-order.
2. Weather influence on route deviation is nonlinear and interacting (e.g., high wind + high wave together matter more than either alone) → tree ensembles and neural nets capture this better than linear/Kalman-only approaches.
3. ETA is a tabular-style regression problem once trajectory context is summarized into features → gradient boosting is fast, interpretable (feature importance), and strong on tabular data, so it's used for ETA rather than forcing a deep sequence model to also do ETA.
4. Route clustering needs no labels (labels don't exist for "normal route") → density-based clustering (DBSCAN) is appropriate since it naturally separates dense shipping lanes from sparse/outlier weather-driven detours.

Performance Parameters

Task	Metric	Why
Trajectory prediction	ADE (Average Displacement Error, in km via Haversine)	Standard metric for trajectory-prediction tasks — average distance between predicted and true path over all timesteps
Trajectory prediction	FDE (Final Displacement Error)	Error specifically at the last predicted point — most relevant for "where will it end up"
Trajectory prediction	RMSE / MAE on lat & lon (or on Haversine distance)	Standard regression error, easy to interpret in km
ETA prediction	MAE, RMSE (in minutes/hours)	Directly interpretable business metric
ETA prediction	R^2 score	Explains variance captured by the model
Route clustering	Silhouette score, number of clusters found	Evaluates cluster quality/lane separation (unsupervised, so no ground-truth accuracy)
Weather-risk flagging (if framed as classification)	Precision, Recall, F1, Confusion Matrix	Because false negatives (missed risk) are costlier than false positives — worth discussing this trade-off explicitly
System	Inference latency, training time	Practical deployability discussion for the "future scope"

Expected Outcomes

1. **Accurate Ship Trajectory**
   - Predict future latitude & longitude
   - Compare predicted vs. actual routes

2. **Improved ETA Prediction**
   - Estimate arrival time using vessel + weather data
   - Compare with non-ML baseline

3. **Weather-Based Route Deviation**
   - Detect deviations from shipping lanes
   - Identify severe-weather affected segments

4. **Interactive Visualization**
   - Display predicted vs. actual routes
   - Show ETA and weather-risk information

5. **Performance Evaluation**
   Trajectory: ADE & FDE
   ETA: MAE, RMSE & R^2

Contribution

Team Member	Contribution
Ronak Shah	Sponsor coordination and system architecture design
Aditya Yadav	Dataset research
Aditya Zalte	Model selection and justification
Aditya Malode	Performance evaluation and explainability framework

Conclusion

- The proposed system combines AIS vessel-tracking data with ocean and weather parameters for intelligent ship-route prediction.
- LSTM/Seq2Seq is proposed for learning sequential vessel movements and predicting future trajectories.
- XGBoost is proposed for ETA prediction using vessel, distance and weather-related features.
- DBSCAN can identify common shipping lanes and detect possible route deviations.
- Performance will be evaluated using ADE, FDE, MAE, RMSE and R^2.
- The system aims to provide better route prediction, ETA estimation and weather-aware route analysis compared with simple movement-based baselines.

References

- NOAA Marine Cadastre — AIS Data (https://marinecadastre.gov/ais/)
- NOAA International Comprehensive Ocean-Atmosphere Data Set (ICOADS) — https://icoads.noaa.gov/
- Kaggle: "AIS Data Complete Weather" — nadaemad2002
- Kaggle: "2024 US Coast Guard + NOAA AIS Dataset" — bwandowando
- Kaggle: "AIS Ship Tracking – Vessel Dynamics and ETA Data" — satyamrajput7913
- IMO — International Maritime Organization, guidance on weather routing and safety of navigation.
- Copernicus Marine Environment Monitoring Service (CMEMS) documentation, for ocean current/wave data.